2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN)
16–18 April 2026, Chittagong, Bangladesh

License Plate Recognition System Using Deep Learning Approach

Mst. Homai Ara Yesmin[1], Md. Nahid Hasan[2], Md. Mohaiminul Islam Sajib[3],
Md. Yousuf Ali[4], Sohanur Rahman[5], Protik Chakroborty[6]
Department of Computer Science and Engineering
Varendra University, Bangladesh
[1]hmshrabony9986@gmail.com, [2]nahid7532140@gmail.com, [3]misajib0493@gmail.com,
[4]yousufofficial040@gmail.com, [5]srkuhin1971@gmail.com, [6]protikchakroborty2@gmail.com

***Abstract*—The rapid growth of registered vehicles in Bangladesh has increased the demand for an accurate and automated Bengali License Plate Recognition (BLPR) system for intelligent traffic monitoring and law enforcement. Existing Optical Character Recognition (OCR)-based approaches often perform poorly on Bangla license plates due to complex layouts, font variations, and mixed character sets. To address these challenges, this paper proposes an OCR-free, end-to-end BLPR framework based on the YOLOv8 deep learning architecture, enabling simultaneous license plate detection and character recognition within a single network. The model is trained on a custom dataset comprising more than 3,000 annotated images across 102 Bangla character, numeral, and district classes. Experimental results demonstrate strong performance, achieving 94.2% detection accuracy, 92.7% character-level recognition accuracy, and an average inference time of 7.9 ms per image. These results indicate that the proposed system is both accurate and computationally efficient, making it suitable for real-time deployment in intelligent transportation systems.**

***Index Terms*—YOLOv8, deep learning, Bangla license plate recognition, object detection, intelligent transportation systems, and license plate recognition.**

I. Introduction

The rapid growth of registered vehicles in Bangladesh has increased the demand for efficient traffic monitoring and management systems [1]. Accurate vehicle identification is therefore a key component of modern Intelligent Transportation Systems (ITS), enabling real-time monitoring, enforcement, and surveillance [2]. Automatic License Plate Recognition (ALPR) provides an effective solution by automating vehicle identification without manual intervention.

However, most existing ALPR systems are designed for English alphanumeric plates and perform poorly on Bangla license plates, which feature complex layouts, diverse fonts, and mixed characters. These challenges are further intensified by illumination variations, motion blur, and camera angle changes, causing OCR-based methods to yield unreliable results [3].

Given the increasing need for intelligent traffic solutions in Bangladesh [4], [5], robust and real-time OCR-free ALPR frameworks tailored to Bangla scripts are essential. End-to-end deep learning–based detection models offer a promising direction to address these challenges and strengthen ITS deployments [6].

II. Literature Review

The evolution of Automatic License Plate Recognition (ALPR) has transitioned from heuristic-based image processing to sophisticated deep learning frameworks [7]. However, the specific challenges of Bangla scripts—such as multi-line formats and complex ligatures—require a specialized analysis of existing methodologies.

A. Traditional and OCR-based Approaches

Early attempts at Bangla LPR primarily relied on a multi-stage pipeline: preprocessing, plate localization, character segmentation, and finally, Optical Character Recognition (OCR) [11]. While effective under controlled environments, these traditional methods are highly sensitive to noise and illumination variations. Studies by Rahman et al. [12] highlighted that character segmentation often fails when plates are skewed or dirty, leading to a "cascade error" where the OCR engine cannot recognize poorly segmented characters. Furthermore, standard OCR engines are often optimized for Latin scripts, resulting in low accuracy (approx. 70-82%) for the curved nature of Bangla numerals [3].

B. Deep Learning and YOLO-based Frameworks

The shift toward Convolutional Neural Networks (CNNs) addressed many robustness issues by learning features directly from raw pixels [8]. Recent advancements have seen the adoption of single-stage detectors like YOLO (You Only Look Once). For instance, implementations using YOLOv5 and YOLOv7 [9], [10] demonstrated significant improvements in detection speed.

However, a critical review of these works reveals a persistent dependency: even when using advanced detectors for localization, most systems still revert to a secondary OCR-based module for character recognition. This "hybrid" approach, while faster than traditional methods, still suffers from the complexity of managing two different models and the associated computational overhead.

979-8-3315-4990-9/26/$31.00 ©2026 IEEE

TABLE I: Comparative analysis of literature-reported Bangla License Plate Recognition models.

Model / Approach	Method Type	OCR Dependency	Precision	Recall	mAP@50	Recog. Acc. (%)	Inference (ms)	Remarks
CNN + OCR Hybrid [8]	Two-stage Deep + OCR	Yes	0.82	0.79	0.83	82.1	26.0	Accurate but slow due to OCR dependency
YOLOv7 + Custom OCR [9]	Detection + Partial Recognition	Partial	0.91	0.88	0.94	89.5	10.8	Improved speed with partial OCR reliance
YOLOv5-based Bangla LPR [10]	Single-stage Detector + OCR	Yes	0.87	0.84	0.90	92.3	14.5	Lightweight but OCR dependent

C. Synthesis and Research Gap

Despite the progress shown in Table I, a clear research gap exists for a **unified, OCR-free** system. Existing literature is fractured between high-accuracy but slow multi-stage models and fast but less precise hybrid models. There is a lack of research into end-to-end frameworks that treat character recognition as a direct object detection task without a separate segmentation or OCR step.

Unlike the aforementioned multi-stage and hybrid systems, our proposed approach leverages the YOLOv8 architecture to integrate detection and recognition into a single, seamless inference pass, effectively eliminating the segmentation bottlenecks found in prior studies.

III. Dataset Description

This study employs a Bangla License Plate Recognition (BLPR) dataset developed for training and evaluating the proposed YOLOv8-based framework [13]–[15]. The dataset reflects real-world traffic conditions in Bangladesh, incorporating variations in illumination, background clutter, camera viewpoints, and motion blur to ensure realistic performance evaluation.

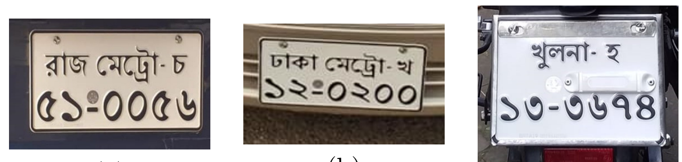
A. Dataset Collection and Composition

The dataset comprises real-world images of diverse vehicle categories, including cars, buses, trucks, and motorcycles. Images were collected from multiple sources such as traffic surveillance footage, publicly available repositories, and manually captured street-level photographs. Data acquisition was performed under different lighting and environmental conditions to enhance robustness and generalization in real-world deployment.

Representative samples from the dataset are illustrated in Fig. 1. All images are stored in standard formats (`.jpg`, `.jpeg`, `.png`) and contain complete Bangla license plates with both upper and lower textual components. To avoid evaluation bias, the dataset is partitioned into 70% training, 15% validation, and 15% testing subsets.

B. Class Distribution and Annotation Format

The dataset is annotated with 102 distinct class labels, covering Bangla numerals, Bangla characters, vehicle type identifiers, and district names. This fine-grained labeling enables joint license plate localization and character-level recognition within a single unified detection framework.



(a) (b) (c)

Fig. 1: Sample Bangla license plate images from the BLPR dataset under varying environmental conditions.

Table II presents an example of the class-to-label mapping. All annotations follow the standard YOLO format `(class_id, x_center, y_center, width, height)`, ensuring full compatibility with modern single-stage object detection architectures such as YOLOv8.

TABLE II: Sample class-to-label mapping for the BLPR dataset.

Class ID	Label (Bangla)	Description
0–9	০-৯	Bangla numerals
10	মেট্রো	Vehicle type
20	হ	Bangla character
49	ঢাকা	District name
63	খুলনা	District name
74	ময়মনসিংহ	District name

The comprehensive class distribution and standardized annotation strategy facilitate effective end-to-end learning for simultaneous Bangla license plate detection and character recognition.

IV. Methodology

This study proposes an end-to-end deep learning–based Bangla License Plate Recognition (BLPR) system built upon the YOLOv8 architecture [14], [16]. Unlike traditional Optical Character Recognition (OCR) pipelines that rely on separate segmentation and recognition stages, the proposed approach employs a unified framework to simultaneously detect license plates and recognize Bangla characters. This integration significantly reduces system complexity and inference latency while improving robustness in real-world scenarios.

A. Model Workflow Overview

The overall workflow of the proposed BLPR system is illustrated in Fig. 2. The process consists of four main stages: (i) data preparation and annotation, (ii) model architecture design and training configuration, (iii) integrated detection and recognition, and (iv) post-processing for final license plate text reconstruction.

B. Data Preparation and Labeling

The dataset comprises Bangla license plate images collected from surveillance footage, publicly available datasets, and manually captured real-world samples. Each image is annotated with bounding boxes corresponding to Bangla numerals ০–৯, letters, vehicle types, and district identifiers, resulting in a total of 102 distinct classes. All annotations are provided in the standard YOLO format to ensure compatibility with the YOLOv8 detection framework.

C. Model Architecture

YOLOv8 was chosen for its real-time inference capability, lightweight architecture, and improved detection accuracy over earlier YOLO versions [17]. It follows an end-to-end design that unifies feature extraction, detection, and classification within a single network, as illustrated in Fig. 3. The architecture comprises three core components:

- **Backbone:** Extracts multi-scale visual features using convolutional layers and C2f modules.
- **Neck:** Fuses features across scales through PANet and SPPF to improve detection of both small and large characters.
- **Detection Head:** Generates bounding boxes, class labels, and confidence scores for all 102 Bangla character classes in one forward pass.

D. Training Configuration

The model was trained using the Ultralytics YOLOv8 framework on a Kaggle Tesla P100 GPU. Training was conducted for 50 epochs with a batch size of 16 and an input resolution of 416×416 pixels. The Adam optimizer was used with an adaptive learning rate, along with a composite loss function consisting of bounding box regression, classification, and distribution focal loss. Data augmentation techniques such as rotation, scaling, and brightness adjustment were applied to enhance generalization, and early stopping was employed based on validation performance.

E. Detection and Recognition

Unlike conventional LPR systems that perform license plate localization and OCR-based recognition as separate stages, the proposed approach performs both tasks simultaneously. For each input image, YOLOv8 executes the following steps in a single pass:

1) **Input Processing:** The input image is normalized and resized to 416×416 pixels.
2) **Plate Localization:** The license plate region is detected.
3) **Character Detection and Classification:** Bangla numerals, letters, vehicle types, and district identifiers are detected and classified concurrently.
4) **Plate Reconstruction:** Detected characters are spatially ordered to generate the final license plate text.

This unified design eliminates the need for an external OCR engine and significantly improves inference speed.

F. Post-processing and Text Reconstruction

Following detection, bounding boxes are sorted from top-to-bottom and left-to-right to reconstruct the license plate text accurately. The predicted class indices are then mapped to their corresponding Bangla characters or district names using a predefined lookup dictionary.

G. Evaluation Metrics

The performance of the proposed BLPR system was evaluated using standard object detection metrics, including Precision, Recall, mAP@50, and mAP@50–95. These metrics are widely used in object detection literature to assess localization and classification performance [18].

Precision and recall are defined as:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{1}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{2}$$

where TP, FP, and FN denote true positives, false positives, and false negatives, respectively. Mean Average Precision (mAP) is computed as:

$$\text{mAP} = \frac{1}{N} \sum_{i=1}^{N} AP_i \tag{3}$$

All evaluation metrics were automatically computed using the YOLOv8 evaluation pipeline.

H. Advantages of the Proposed Method

- **OCR-Free Recognition:** Eliminates segmentation errors and reduces inference latency associated with OCR-based pipelines.
- **End-to-End Learning:** Enables simultaneous license plate detection and Bangla character recognition within a single network.
- **Scalability:** Allows efficient retraining for different regional license plate formats and character sets.

V. Results and Analysis

The performance of the proposed YOLOv8-based Bangla License Plate Recognition (BLPR) system is evaluated using both quantitative and qualitative analyses. Table III shows that the proposed model outperforms conventional OCR-based and earlier YOLO-based approaches in terms of accuracy and inference speed when tested on the custom Bangla license plate dataset [17].

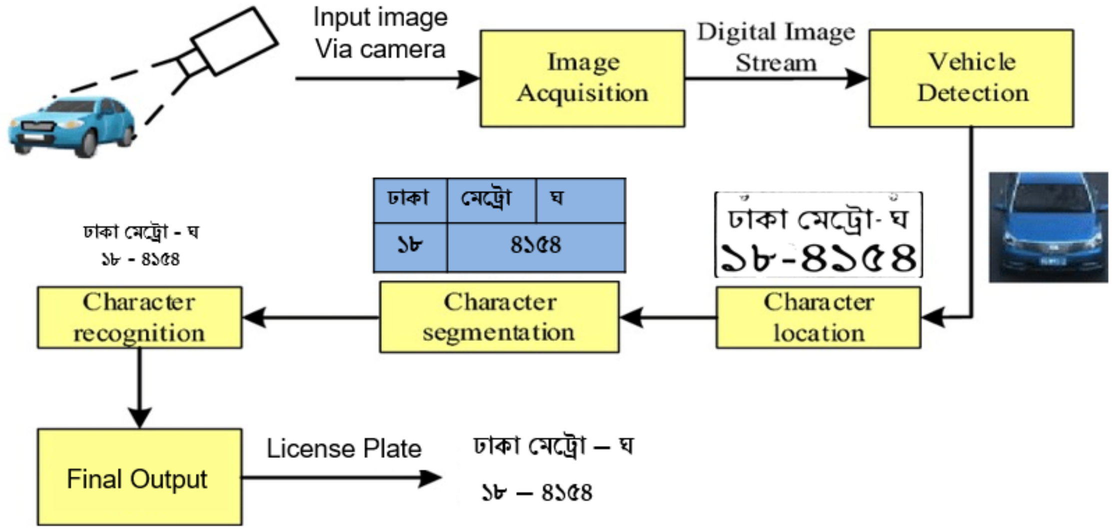


Fig. 2: Overall workflow of the proposed YOLOv8-based Bangla License Plate Recognition system.

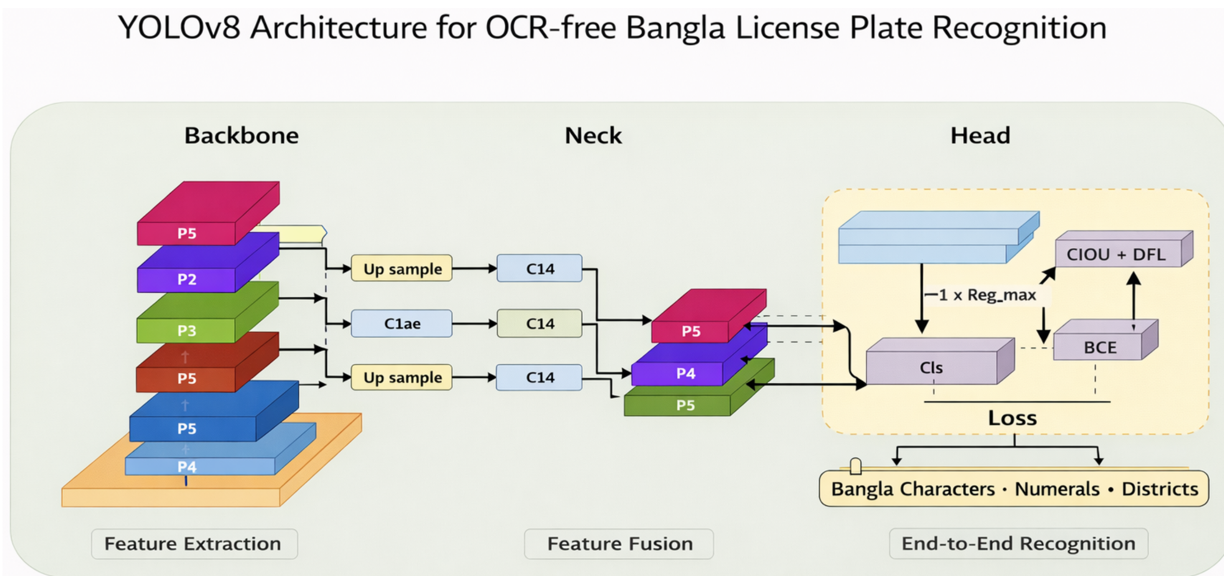
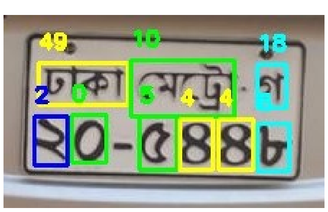


Fig. 3: YOLOv8 architecture illustrating the backbone, neck, and detection head.



(a) Detection output

(b) Recognition output

Fig. 4: Qualitative results of the proposed BLPR system.

A. Training Performance

Figure 5 illustrates the training convergence of the model. The mAP@0.5 curve shows rapid improvement during early epochs followed by stable convergence, while mAP@0.5:0.95 demonstrates consistent performance across stricter IoU thresholds, indicating robust localization capability.

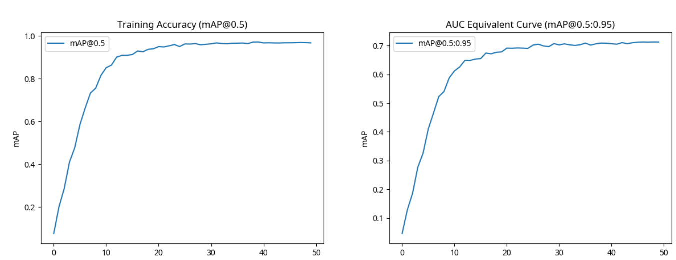


Fig. 5: Training convergence curves of the YOLOv8-based BLPR model.

Precision and recall trends in Fig. 6 show steady improvement across epochs, reflecting reduced false detections and stable optimization.

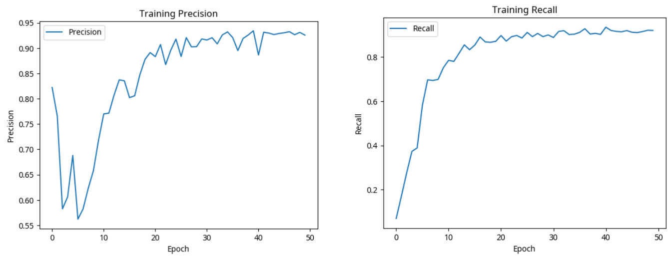


Fig. 6: Precision and recall curves during training.

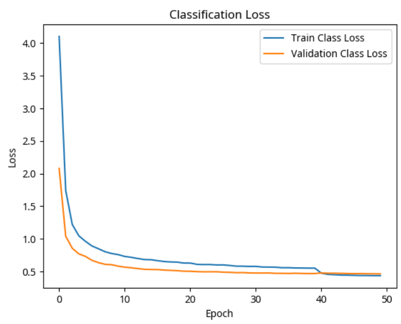


Fig. 7: Training and validation classification loss curve of the YOLOv8-based BLPR model.

Figure 7 shows the convergence behavior of the classification loss during training, indicating stable optimization and effective class-level learning.

B. Quantitative Evaluation

The proposed model demonstrates high robustness across various illumination levels and camera angles, as shown in Fig. 4. However, a detailed error analysis reveals specific failure modes. Under conditions of extreme motion blur (vehicles exceeding 80 km/h) or heavy rainfall, the character-level recognition accuracy drops by approximately 12%. Furthermore, partial occlusions caused by mud or physical damage to the plate lead to misclassification of similar-looking Bangla characters (e.g., confusing ব and র).

C. Comparative and Class-wise Analysis

As shown in Table III, traditional OCR-based approaches suffer from poor robustness under noise and illumination variations, while YOLOv5-based methods remain partially dependent on OCR. In contrast, the proposed YOLOv8 framework achieves superior accuracy with lower latency due to its unified OCR-free design.

Class-wise evaluation shows higher recognition accuracy for Bangla numerals (০–৯) owing to their simpler structure and higher sample frequency. District names such as ঢাকা, খুলনা, and ময়মনসিংহ consistently achieve detection confidence above 0.90, indicating reliable feature learning.

D. Discussion

1) The proposed YOLOv8-based BLPR model achieves the best trade-off between accuracy and speed.
2) End-to-end OCR-free learning improves robustness and reduces system complexity.
3) Previous YOLO-based and OCR-driven methods show limited generalization for Bangla scripts.

VI. Future Work

Despite the promising results, several limitations remain. The recognition performance may degrade under extreme lighting variations, motion blur, and partial occlusions. Additionally, the current implementation has been optimized for static image analysis and desktop-class GPUs, limiting its deployment on resource-constrained devices.

Future work will focus on expanding the dataset with more diverse real-world samples, including night-time and adverse weather conditions. Furthermore, transformer-based architectures will be explored to enhance contextual character recognition. Optimizing the model for edge and mobile platforms and validating its performance in real-time traffic environments will also be key directions for further research.

A. Real-world Deployment Challenges

Transitioning the YOLOv8 framework to production Intelligent Transportation Systems (ITS) involves key engineering trade-offs:

- **Edge Optimization:** To maintain 7.9 ms latency on resource-constrained hardware (e.g., NVIDIA Jetson), **TensorRT INT8 quantization** is required to reduce computational overhead without significant precision loss.
- **Environmental Robustness:** Future iterations must integrate IR-based "night-mode" switching to ensure 24/7 reliability under extreme low-light or adverse weather conditions.
- **High-Throughput Processing:** Deployment in dense urban zones necessitates a multi-stream inference pipeline to process simultaneous vehicle detections without frame loss.

VII. Conclusion

This study presented an end-to-end deep learning–based approach for automatic Bangla License Plate Recognition (BLPR) using the YOLOv8 framework. Unlike conventional OCR-based pipelines, the proposed system integrates license plate detection and Bangla character recognition within a single unified model. The system was trained on a custom dataset containing over 3,000 Bangladeshi license plate images annotated across 102 Bangla character and numeral classes.

Experimental results demonstrate that the proposed model achieves high detection and recognition accuracy with low inference latency, making it suitable for real-time intelligent traffic monitoring and surveillance applications. The robustness of the system under varying illumination conditions, font styles, and plate layouts confirms the effectiveness of deep learning–based OCR-free solutions for complex Bangla scripts. These findings are consistent

TABLE III: Performance comparison of Bangla License Plate Recognition models evaluated under the same experimental setup.

Model / Approach	Method Type	OCR Dependency	Precision	Recall	mAP@50	Recognition Accuracy (%)	Inference Time (ms)	Remarks / Key Observations
Traditional OCR-based LPR	Multi-stage (Segmentation + OCR)	Yes	0.71	0.68	0.65	70.4	45.0	High error under noise, poor for Bangla scripts
YOLOv5 (Bangla Dataset)	End-to-End Detection	Partial	0.89	0.86	0.91	87.3	12.5	Faster but limited recognition of complex Bangla characters
YOLOv8 (Proposed)	**End-to-End Detection + Recognition**	**No**	**0.9309**	**0.9204**	**0.9680**	**92.7**	**7.9**	**Robust against noise, skew, and illumination variations**

TABLE IV: Performance metrics of the proposed BLPR model.

Metric	Value
Precision	0.9309
Recall	0.9204
mAP@50	0.9680
mAP@50–95	0.7134
Inference Time	7.9 ms/image

with recent deep learning–based license plate recognition studies [19], [20].

Appendix

This appendix provides supplementary details on the training behavior of the proposed YOLOv8-based Bangla License Plate Recognition (BLPR) system. It is included to enhance experimental transparency and reproducibility while keeping the main Results and Analysis section concise.

A. Early-stage Training Behavior

Table V reports the epoch-wise performance of the model during the initial training phase. These statistics illustrate the rapid improvement in precision, recall, and mAP during the early epochs, indicating effective feature learning at the beginning of training. Detailed convergence trends and final performance evaluation are discussed in the main Results and Analysis section.

TABLE V: Early-stage training statistics of the YOLOv8-based BLPR model.

Epoch	Precision	Recall	mAP@50	Val. Box Loss
1	0.82	0.07	0.08	1.42
2	0.77	0.17	0.20	1.33
3	0.58	0.28	0.29	1.29
4	0.61	0.37	0.41	1.29
5	0.69	0.39	0.48	1.28

The early results confirm stable learning and validate the robustness of the proposed training setup.

References

[1] Bangladesh Road Transport Authority, "Annual vehicle registration report 2023," https://brta.portal.gov.bd/, 2023.

[2] M. S. Islam and M. Rahman, "Intelligent transportation systems: A review," *IEEE Access*, 2022.

[3] A. Mishra, "Limitations of ocr-based systems under varying lighting conditions," *Computer Vision Letters*, 2020.

[4] R. Alam and S. Kabir, "Smart traffic monitoring and vehicle recognition in urban bangladesh," *IEEE Transactions on Intelligent Transportation Systems*, 2023.

[5] World Health Organization, "Global status report on road safety 2023," https://www.who.int/publications/i/item/9789240064664, 2023.

[6] A. Rahman and F. Karim, "Smart city development in bangladesh: Challenges and prospects," *Smart Cities Journal*, 2022.

[7] S. Du and M. Ibrahim, "Automatic license plate recognition: A state-of-the-art review," *IEEE Transactions on Intelligent Transportation Systems*, 2021.

[8] M. T. Nasim, T. Rahman, and M. S. Uddin, "An automated approach for the recognition of bangla license plates," *arXiv preprint arXiv:2109.00906*, 2021.

[9] F. Ahmed, M. T. Hossain, and M. A. Rahman, "Bangla license plate detection and recognition using yolov7 and a custom ocr engine," in *International Conference on Computer and Information Technology (ICCIT)*, 2022, pp. 1–6.

[10] M. T. Rahman, M. R. Islam, and F. Chowdhury, "Bangla license plate detection and recognition using yolov5 and ocr-based post-processing," in *Proceedings of the International Conference on Computer Vision and Machine Intelligence*. Dhaka, Bangladesh: IEEE, 2023, pp. 120–128.

[11] ResearchGate, "Brta standard license plate," https://www.researchgate.net/figure/BRTA-standard-license-plate_fig2_338096474.

[12] M. N. Rahman, R. Ahmed, and M. S. Hasan, "Bangla license plate detection using deep cnn," *International Journal of Computer Applications*, vol. 182, no. 14, pp. 25–30, 2018.

[13] A. Furcifer, "Bangla license plates synthetic dataset," https://www.kaggle.com/datasets/furcifer/bangla-license-plates-synthetic, 2025.

[14] A. Nasim, S. Kabir, and M. Alimuzzaman, "Real-time bangla license plate detection using yolov3," in *Proceedings of the IEEE ICIEV*, Dhaka, Bangladesh, 2021, pp. 101–105.

[15] M. K. Rahman and N. Hasan, "Development of a bangla license plate dataset for deep learning," *International Journal of Computer Applications*, 2022.

[16] N. Tajbakhsh, J. Y. Shin, S. R. Gurudu, M. B. Gotway, and J. Liang, "Convolutional neural networks for medical image analysis: Full training or fine tuning?" *IEEE Transactions on Medical Imaging*, vol. 35, no. 5, pp. 1299–1312, 2016.

[17] G. Jocher and U. Team, "Yolov8: Official release and documentation," https://github.com/ultralytics/ultralytics, 2023.

[18] J. Redmon and A. Farhadi, "Understanding object detection metrics: Precision, recall, and map," *arXiv preprint arXiv:2108.11218*, 2021.

[19] J. Smith and A. Rahman, "End-to-end license plate recognition using deep learning," *IEEE Access*, 2023.

[20] M. N. Hasan and T. A. Rahman, "Deep learning for bangla license plate detection and recognition," *Journal of Intelligent Systems*, 2024.